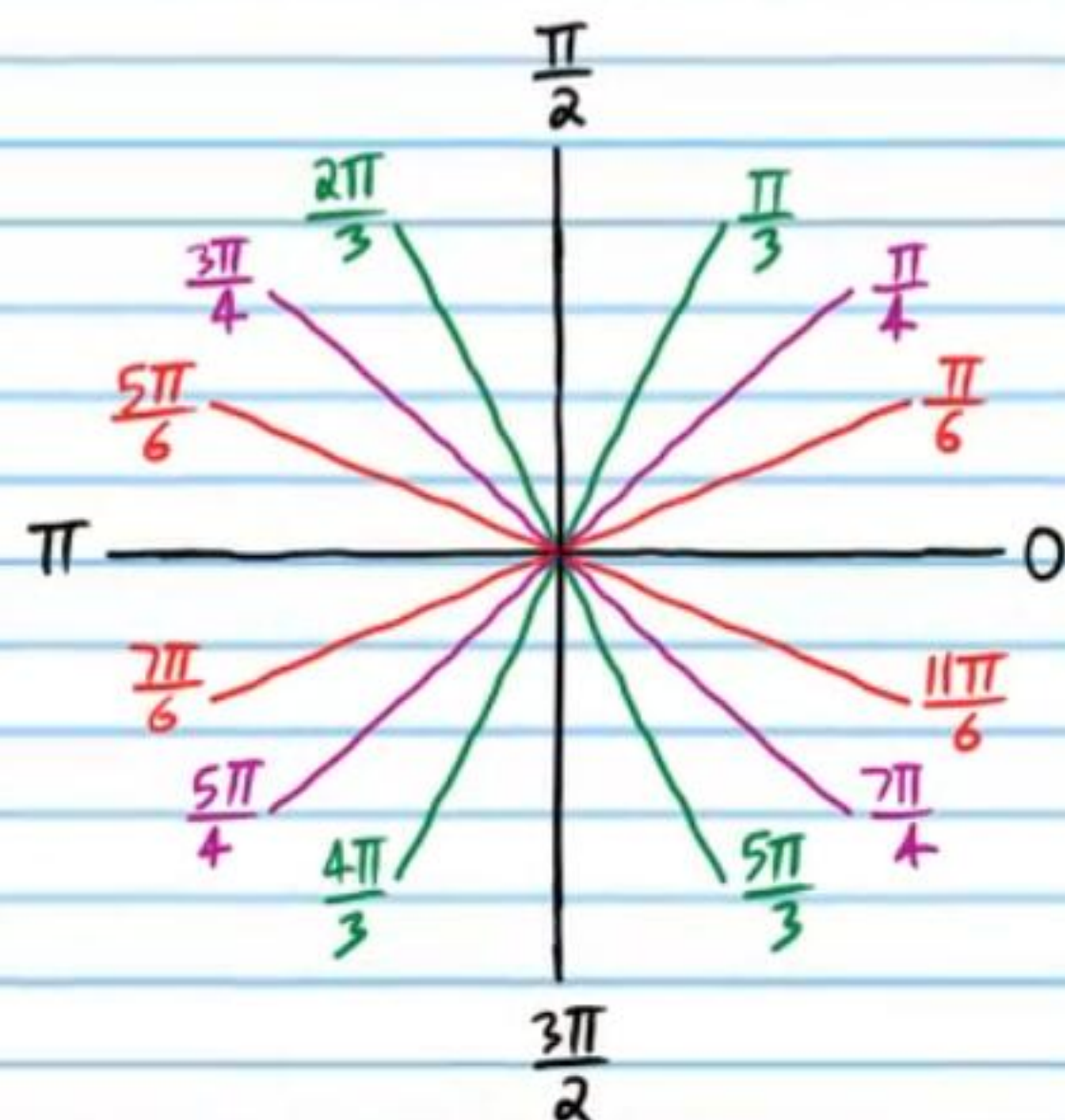
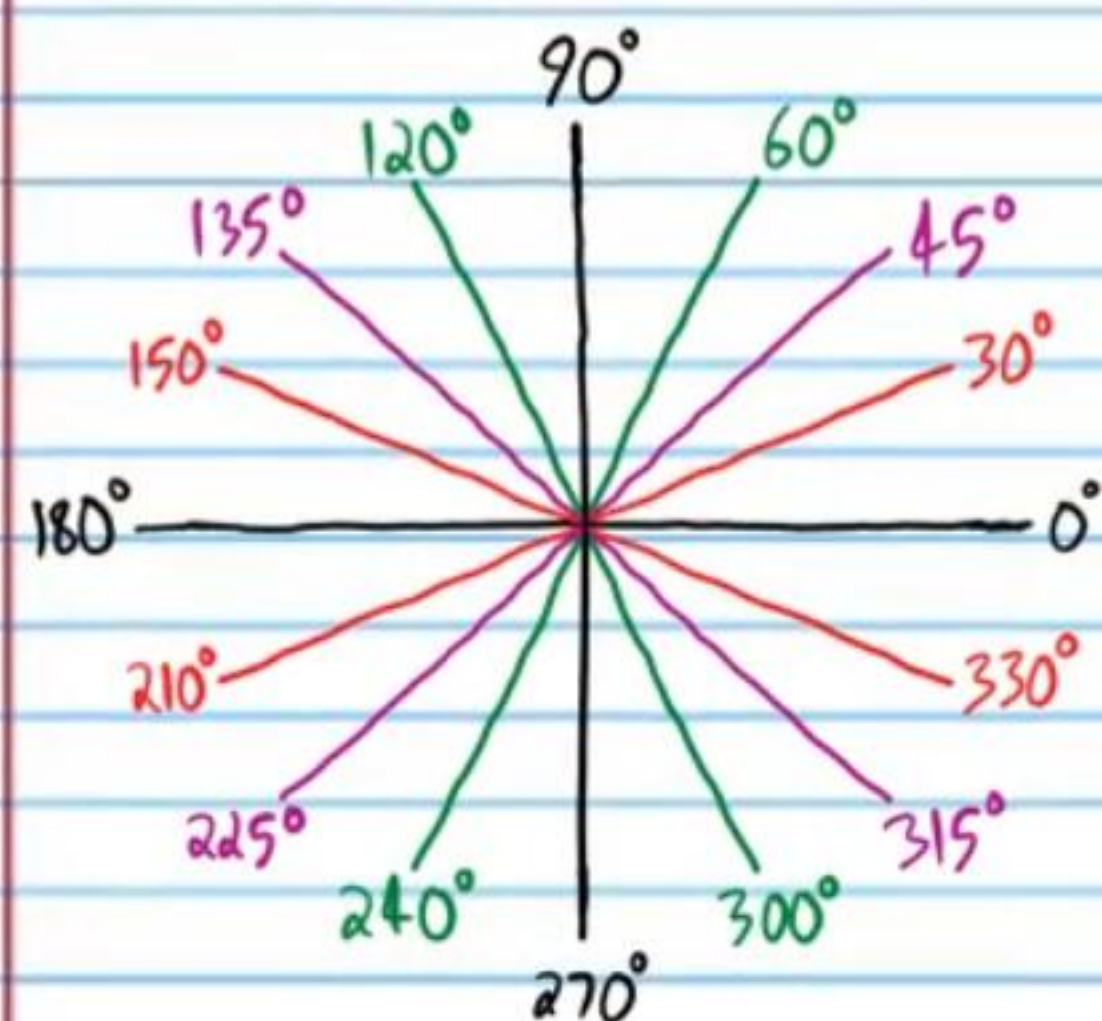


The Complex Plane and Graphing Polar Equations Using Radian Units

Radian Units

Degree Units

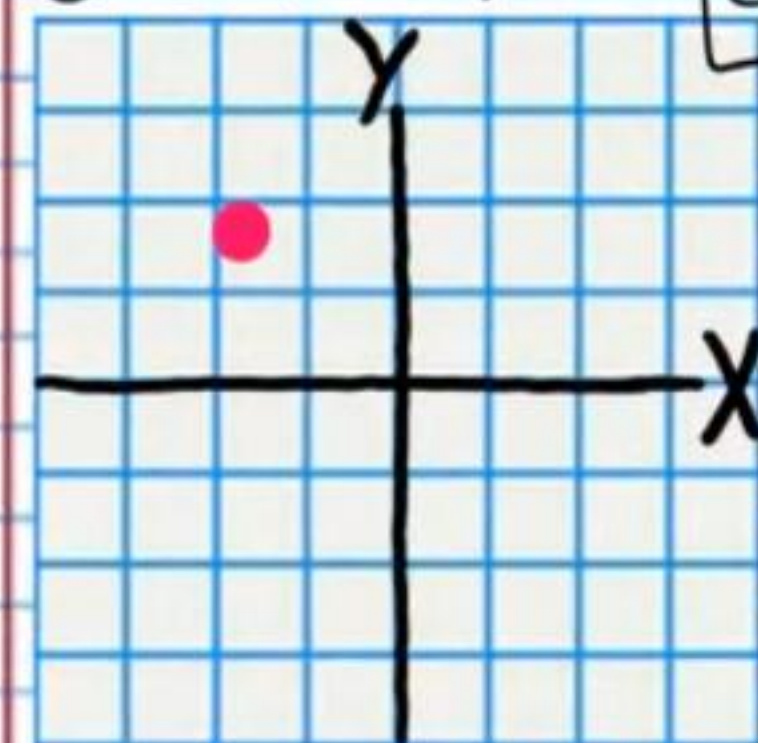
Radian Units



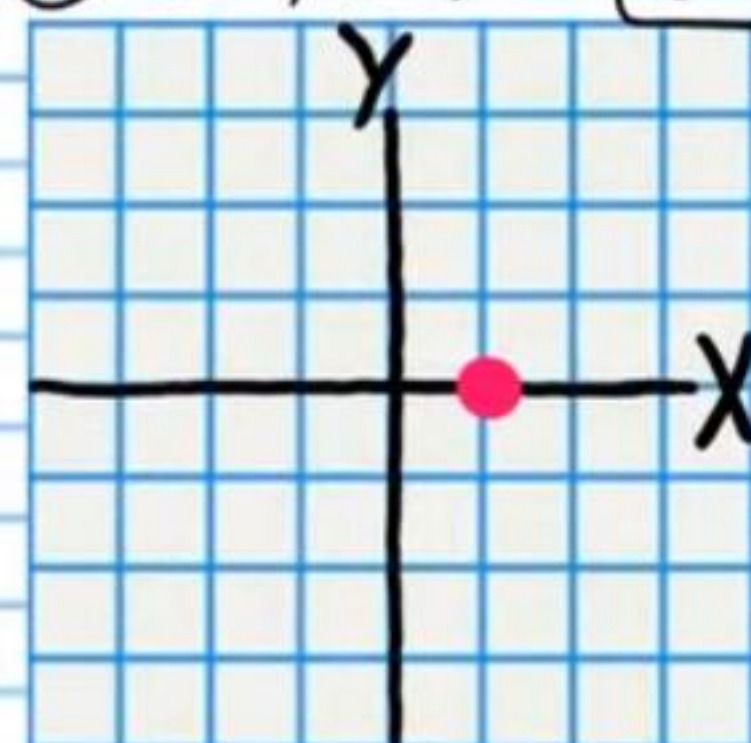
Using Radian Units with Polar Coordinates

Draw the points described by the polar coordinates below. Then convert the coordinates so that each angle measure is written with radian units rather than degree units.

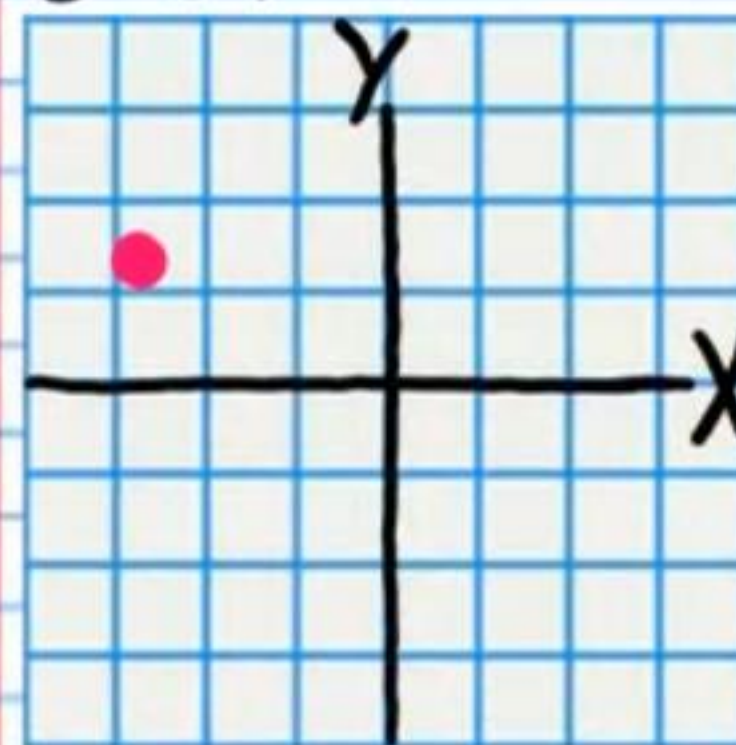
① $(2, 135^\circ) = \boxed{(2, \frac{3\pi}{4})}$



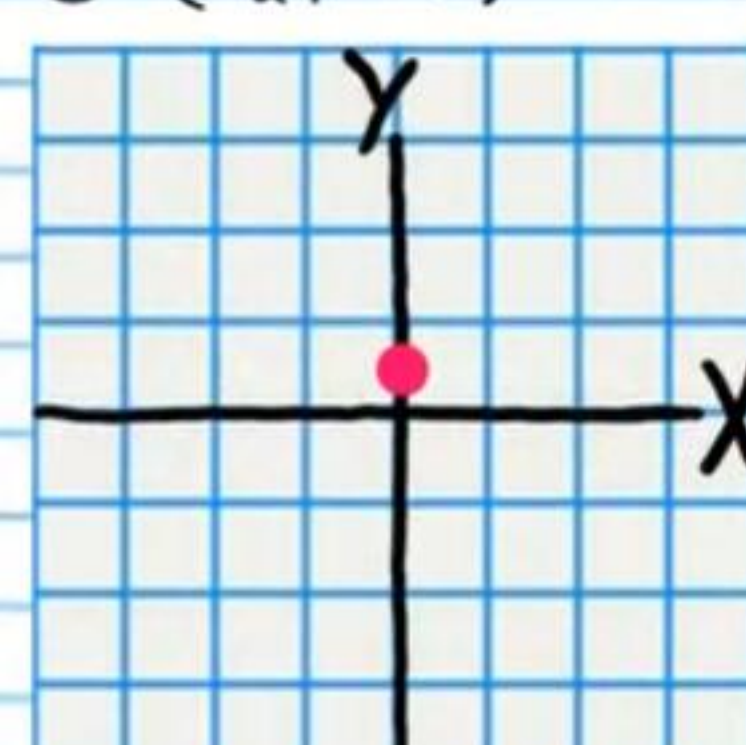
② $(-1, 180^\circ) = \boxed{(-1, \pi)}$



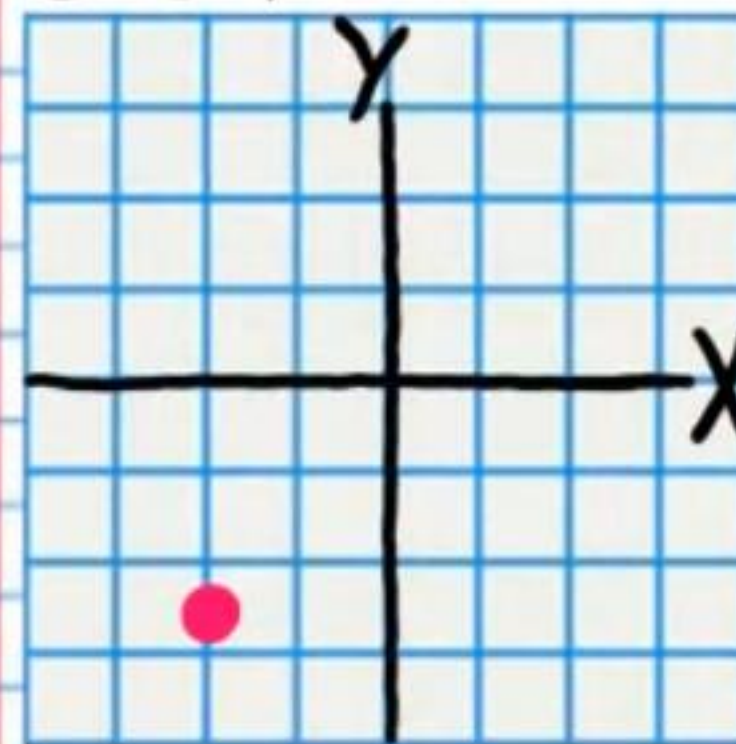
③ $(3, -210^\circ) = \boxed{(3, -\frac{7\pi}{6})}$



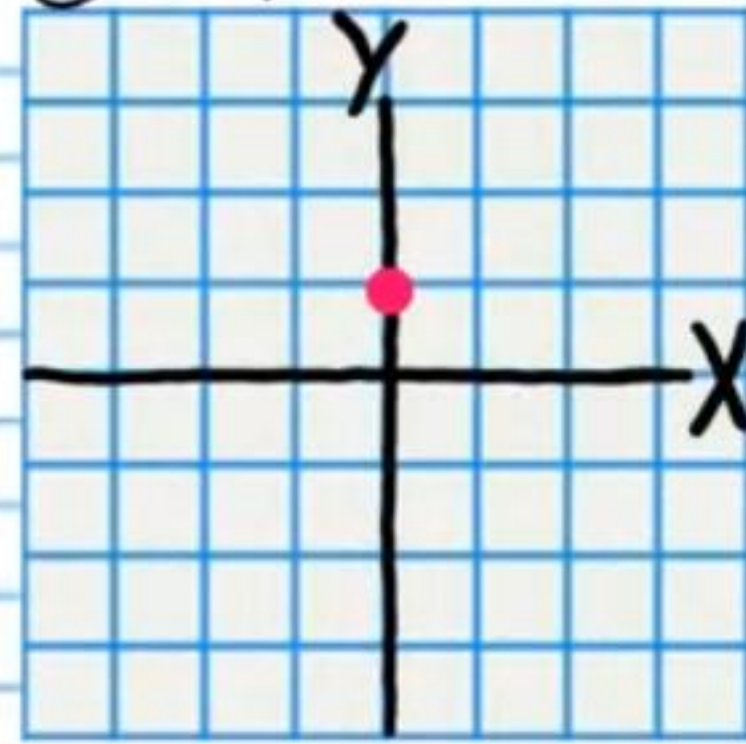
④ $(-\frac{1}{2}, -90^\circ) = \boxed{(-\frac{1}{2}, -\frac{\pi}{2})}$



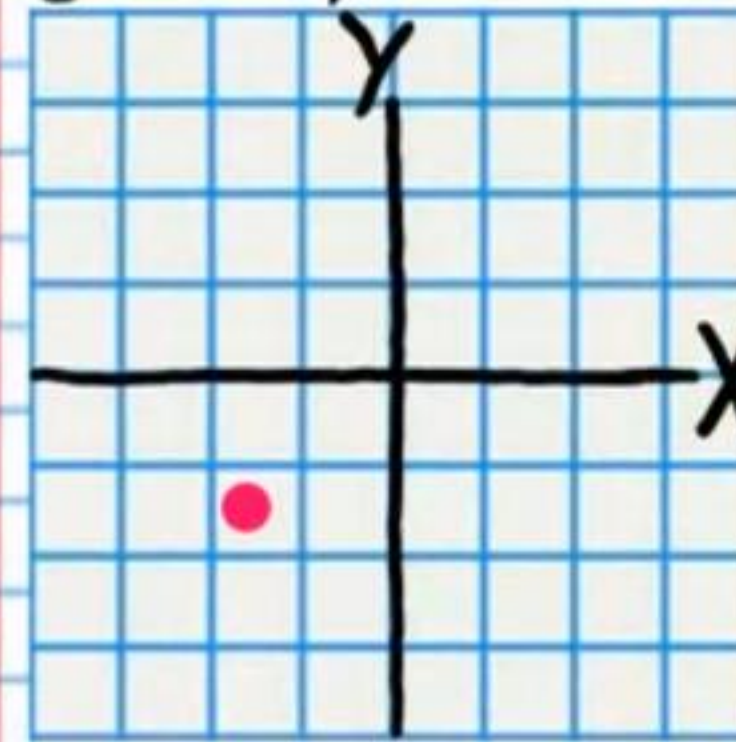
⑤ $(3, 240^\circ) = \boxed{(3, \frac{4\pi}{3})}$



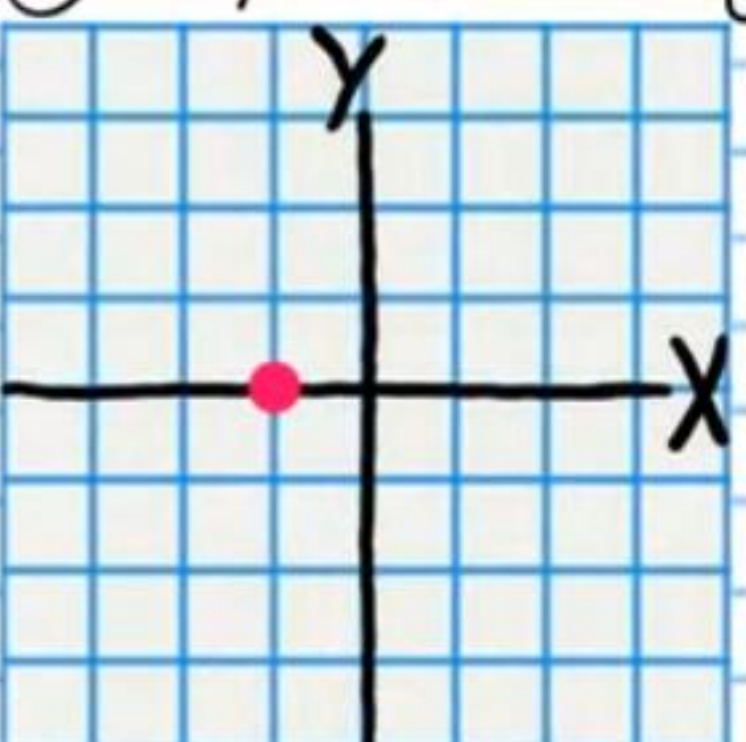
⑥ $(-1, 270^\circ) = \boxed{(-1, \frac{3\pi}{2})}$



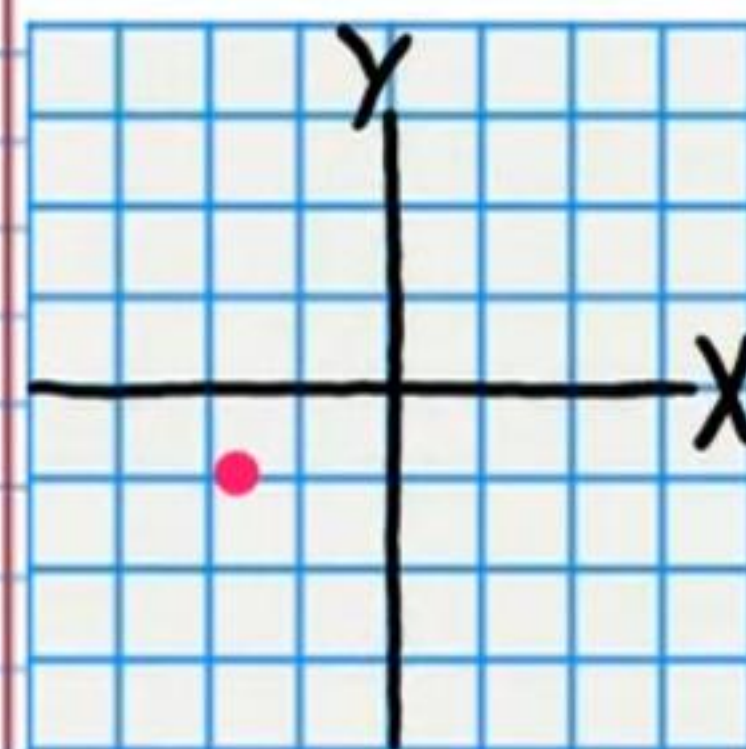
⑦ $(-2, 45^\circ) = \boxed{(-2, \frac{\pi}{4})}$



⑧ $(1, -180^\circ) = \boxed{(1, -\pi)}$

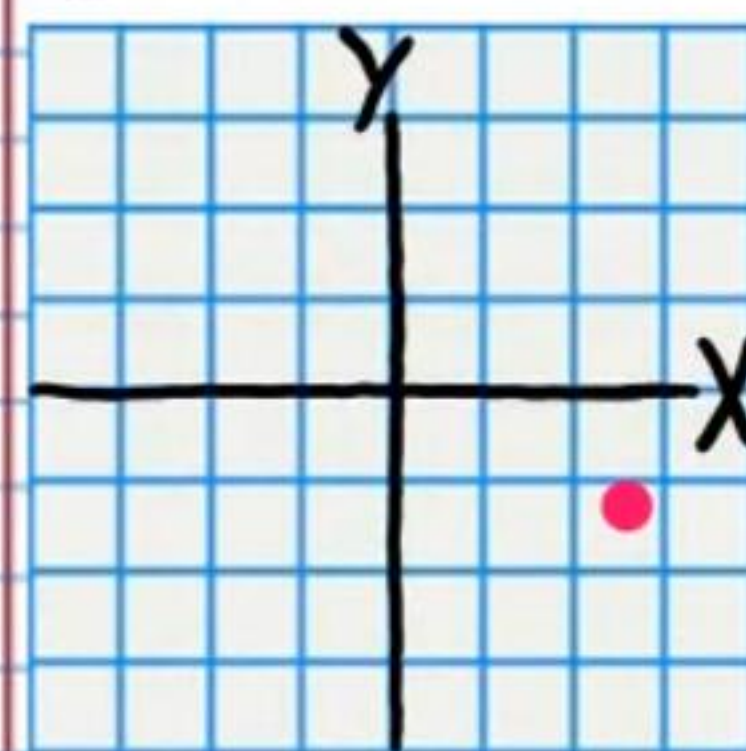


• ⑨ $(2, 200^\circ) = \boxed{(2, \frac{10\pi}{9})}$



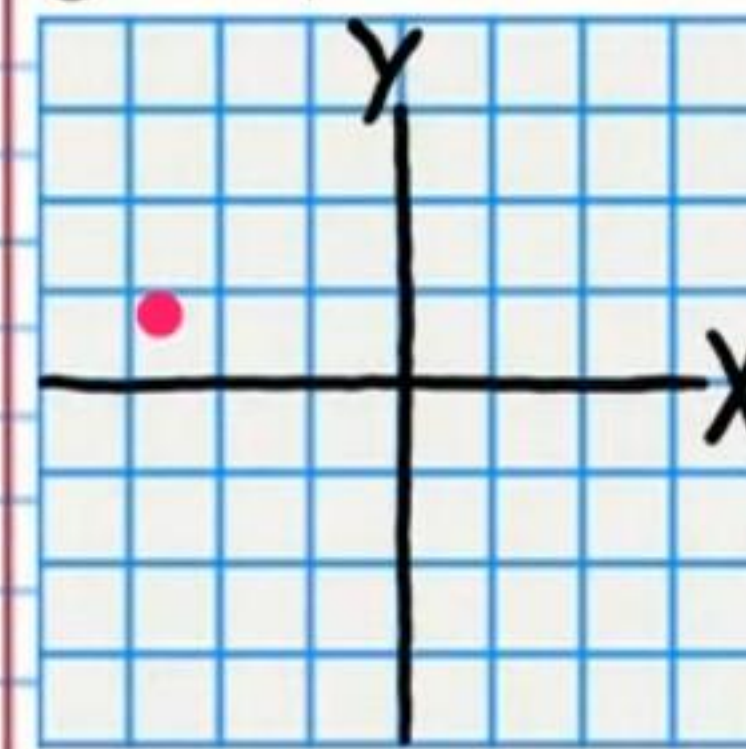
$$200^\circ \times \frac{\pi}{180^\circ} = \frac{200\pi}{180} = \frac{20\pi}{18} = \frac{10\pi}{9}$$

• ⑩ $(-3, -211^\circ) = \boxed{(-3, -\frac{211\pi}{180})}$



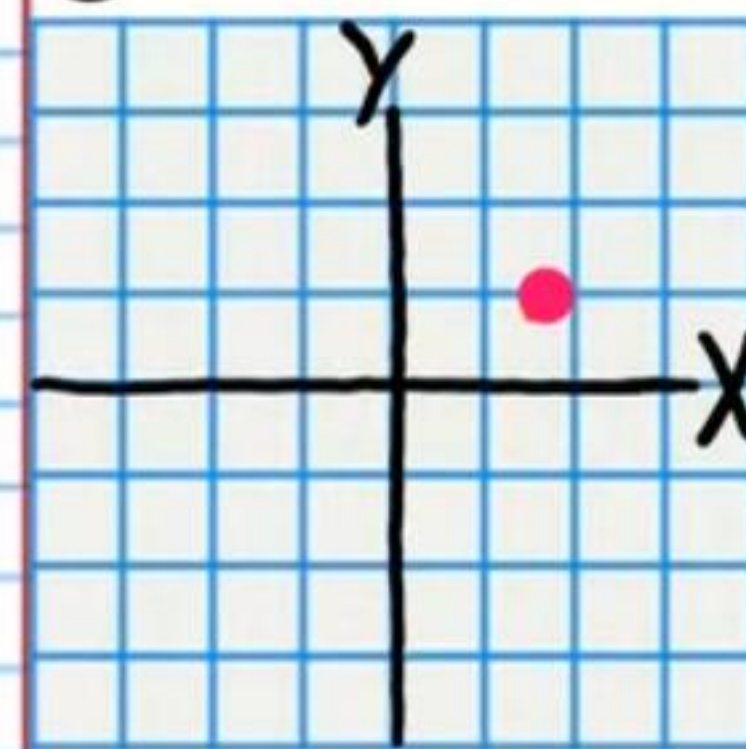
$$-211^\circ \times \frac{\pi}{180^\circ} = -\frac{211\pi}{180}$$

• ⑪ $(-3, -20^\circ) = \boxed{(-3, -\frac{\pi}{9})}$



$$-20^\circ \times \frac{\pi}{180^\circ} = -\frac{20\pi}{180} = -\frac{\pi}{9}$$

• ⑫ $(2, 35^\circ) = \boxed{(2, \frac{7\pi}{36})}$

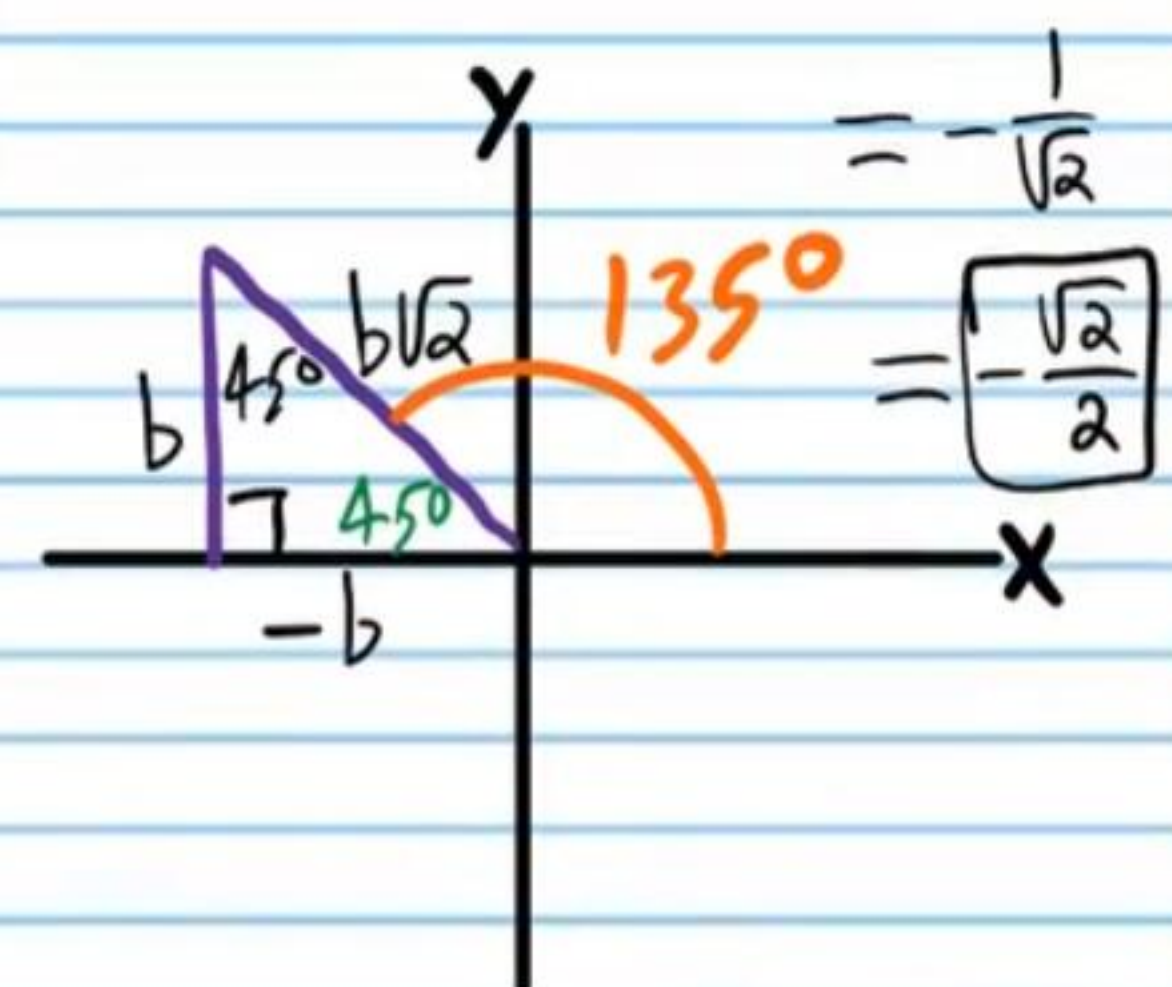
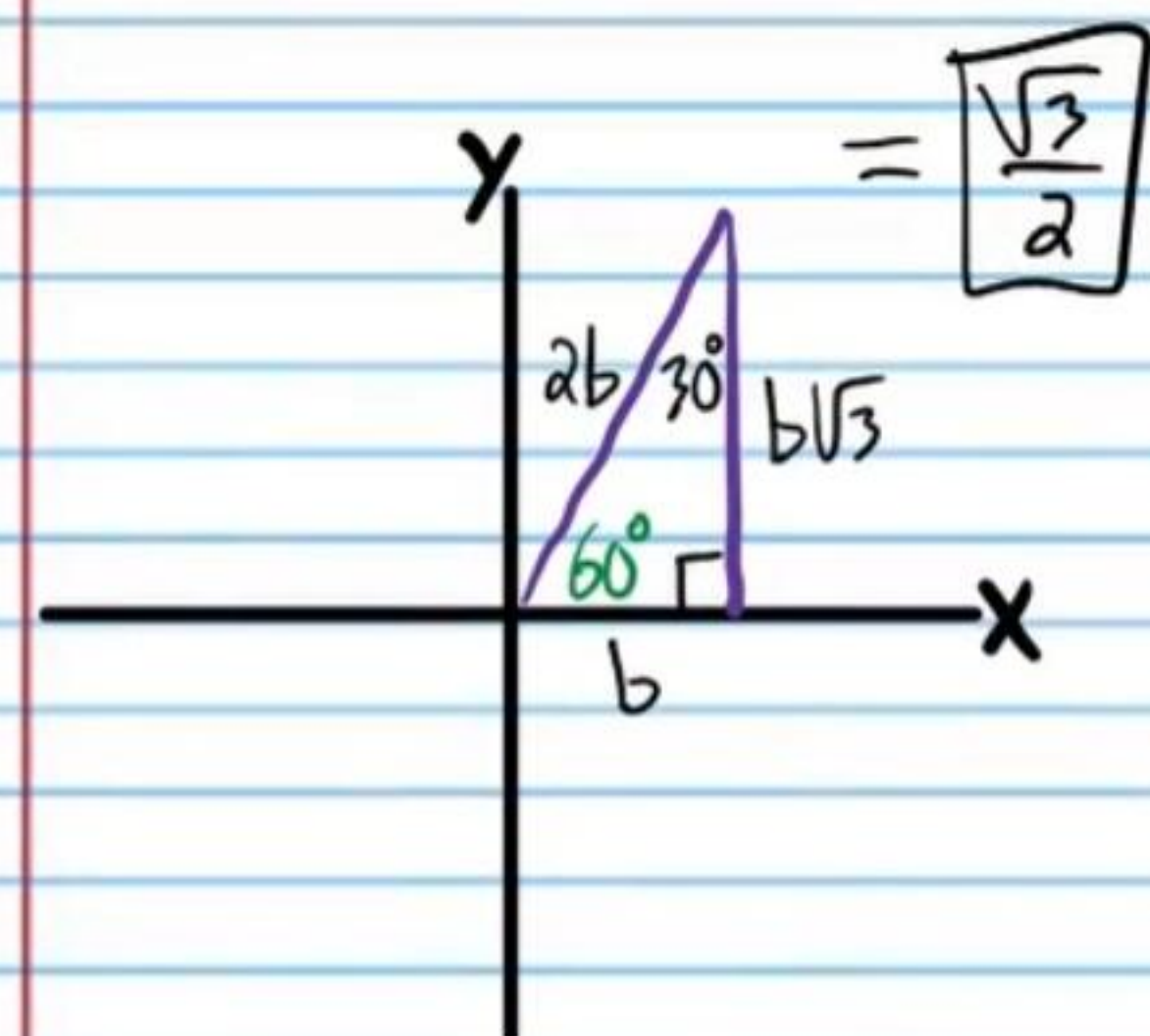


$$35^\circ \times \frac{\pi}{180^\circ} = \frac{35\pi}{180} = \frac{7\pi}{36}$$

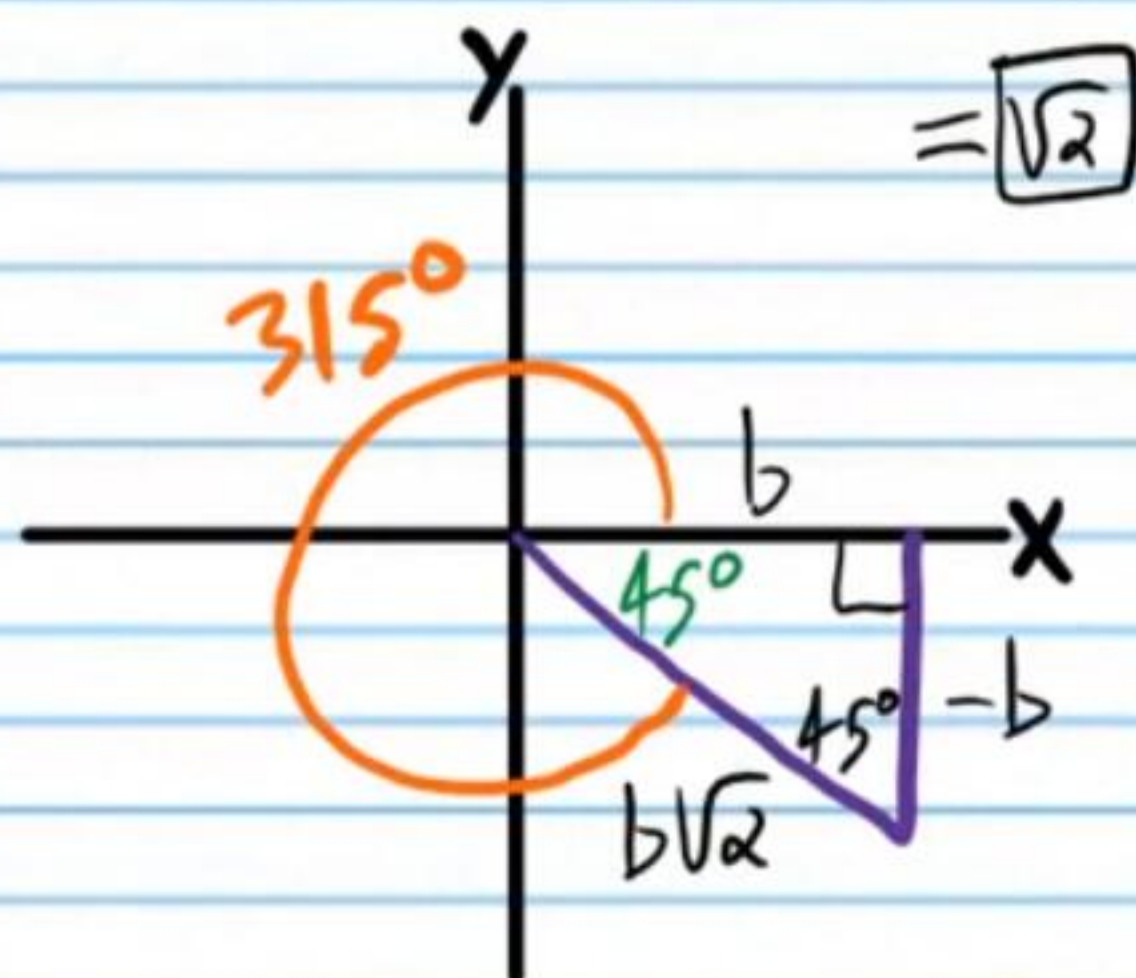
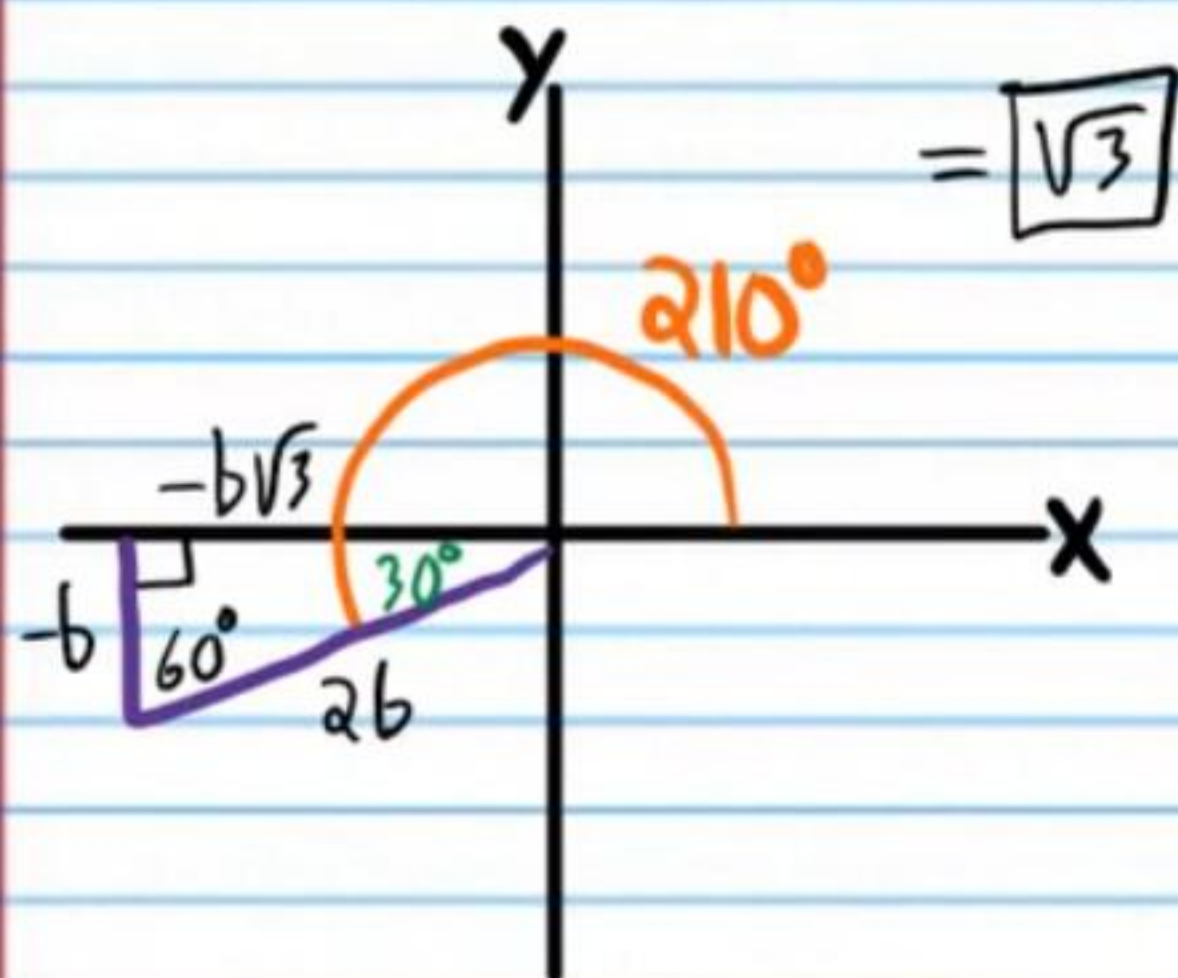
Reviewing Trigonometry with Radian Units

Find the values of the expressions below **without a calculator**.

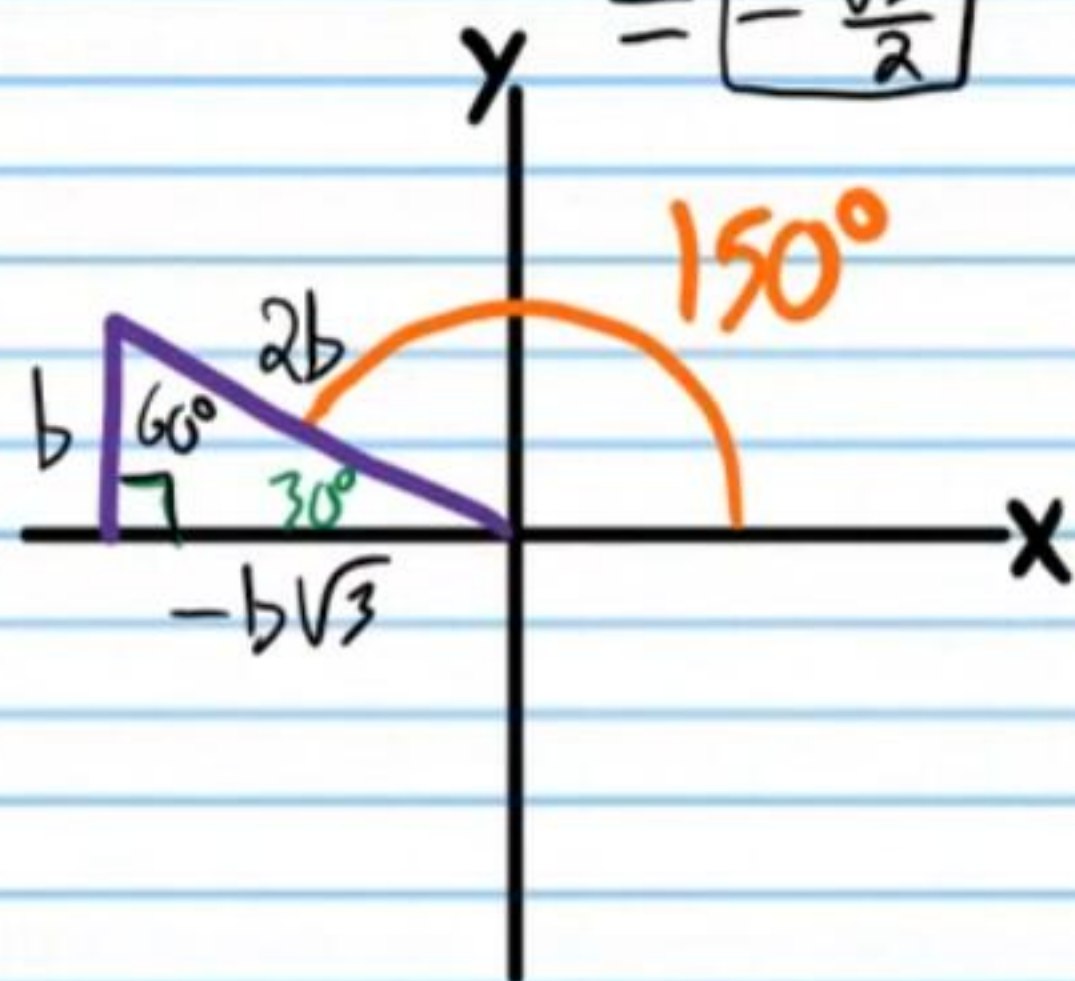
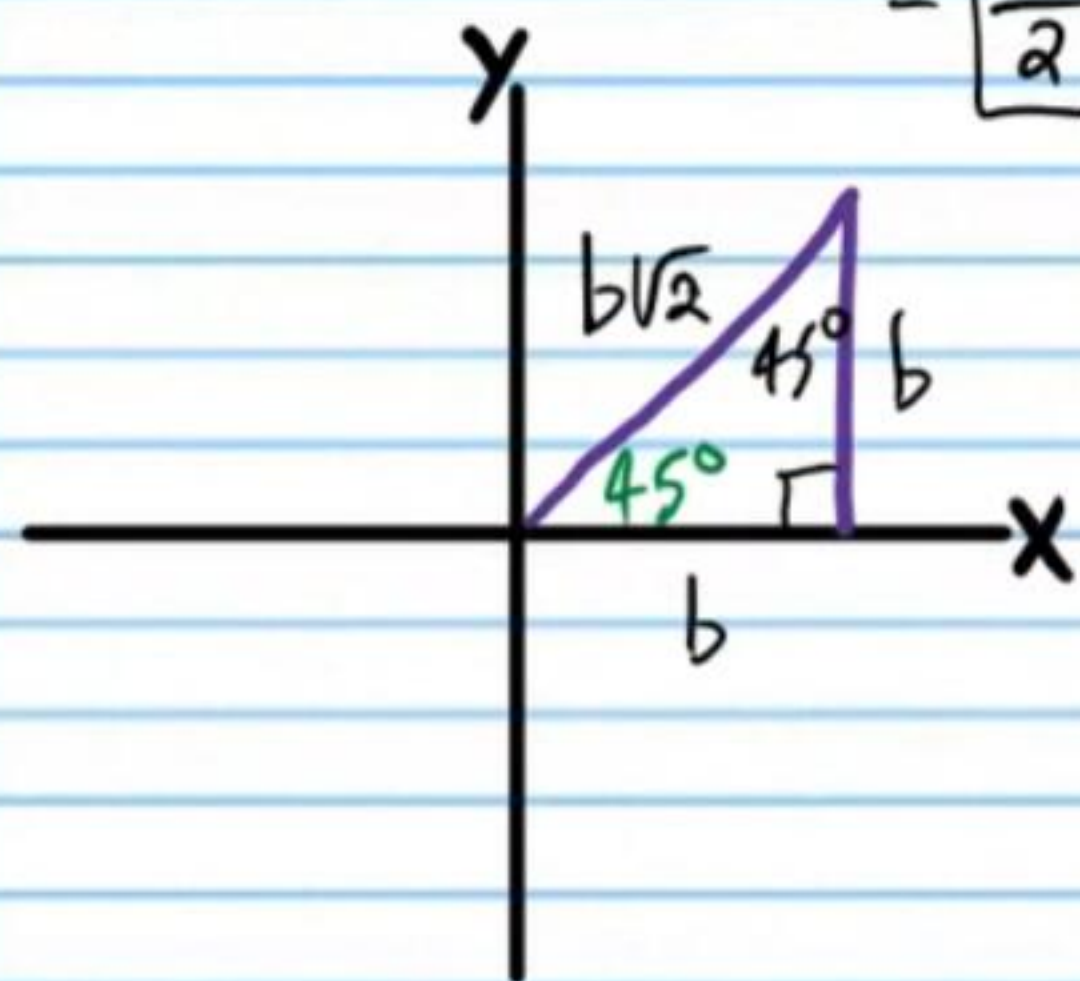
• ⑬ $\sin(\frac{\pi}{3}) = \sin 60^\circ = \frac{b\sqrt{3}}{2b}$ ⑭ $\cos(\frac{3\pi}{4}) = \cos 135^\circ = \frac{-b}{b\sqrt{2}}$



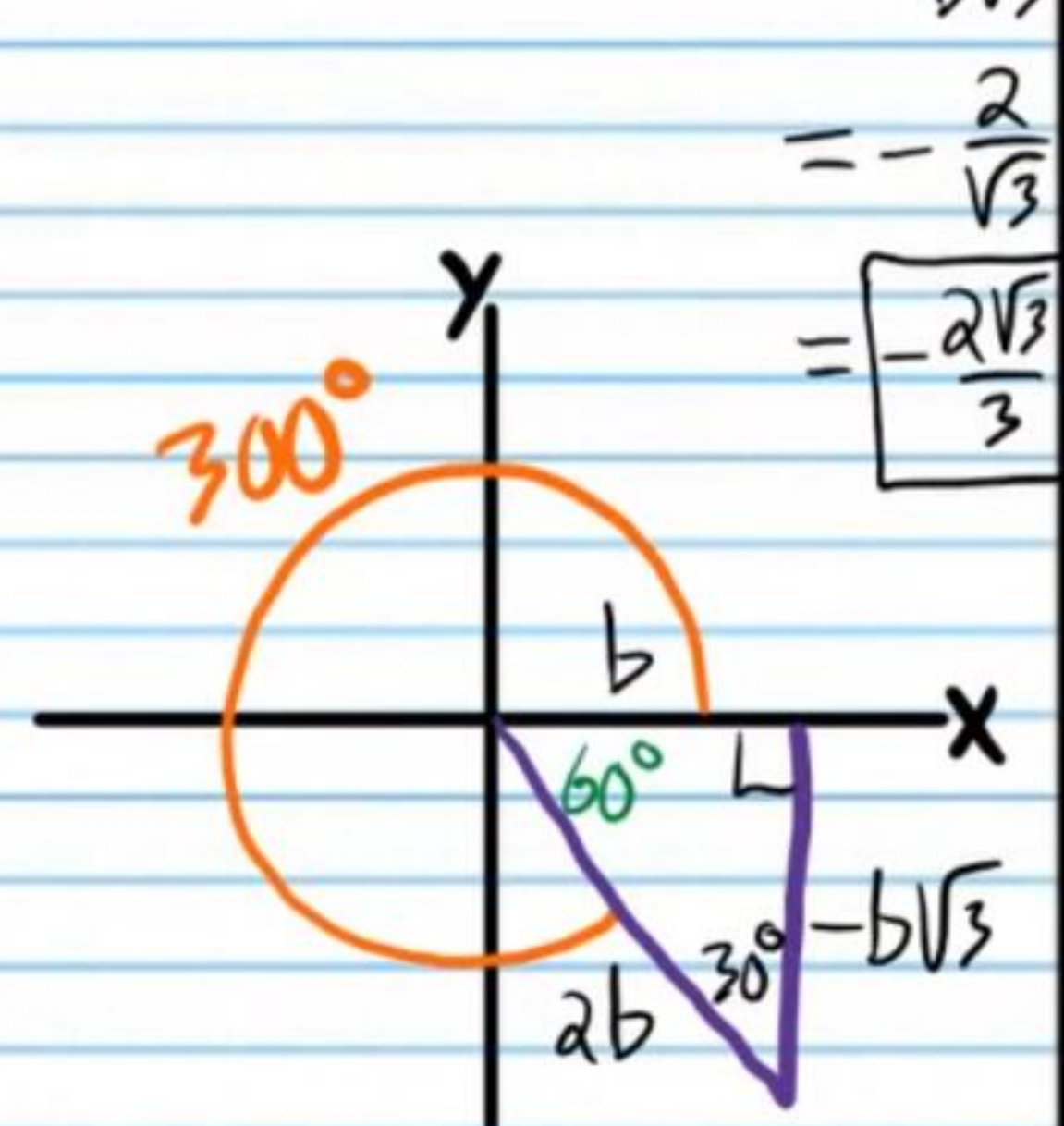
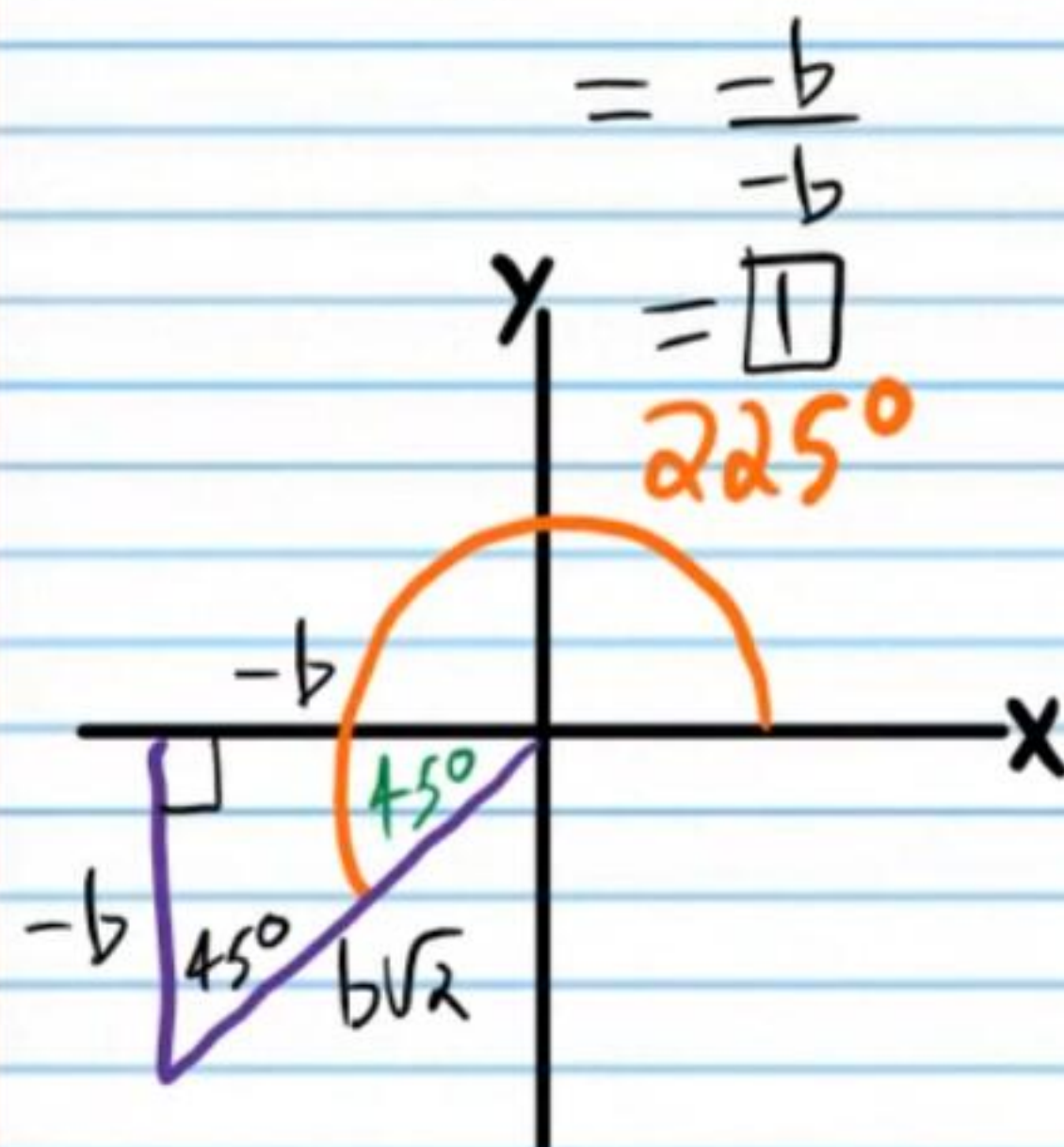
⑮ $\cot\left(\frac{7\pi}{6}\right) = \cot(210^\circ) = \frac{-b\sqrt{3}}{-b} = \boxed{\sqrt{3}}$ ⑯ $\sec\left(\frac{7\pi}{4}\right) = \sec(315^\circ) = \frac{b\sqrt{2}}{b} = \boxed{\sqrt{2}}$



⑰ $\sin\left(\frac{\pi}{4}\right) = \sin 45^\circ = \frac{b}{b\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$ ⑱ $\cos\left(\frac{5\pi}{6}\right) = \cos 150^\circ = \frac{-b\sqrt{3}}{2b} = \boxed{-\frac{\sqrt{3}}{2}}$



⑲ $\tan\left(\frac{5\pi}{4}\right) = \tan 225^\circ = \frac{-b}{-b} = \boxed{1}$ ⑳ $\csc\left(\frac{5\pi}{3}\right) = \csc 300^\circ = \frac{2b}{-b\sqrt{3}} = \boxed{-\frac{2}{\sqrt{3}}}$



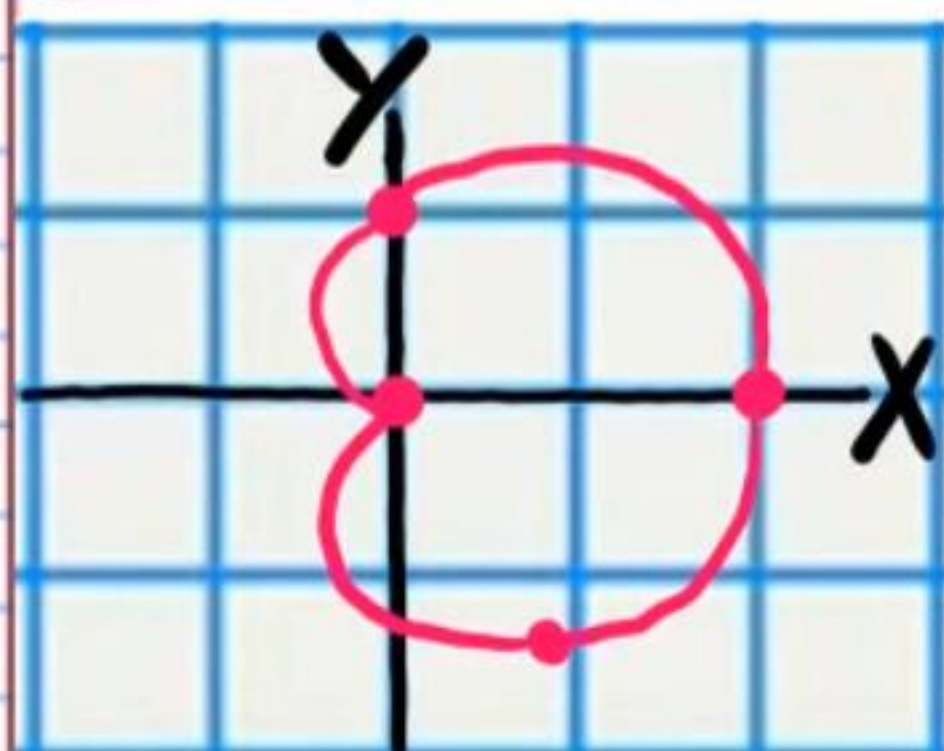
②① $\sin \pi = \sin 180^\circ = \boxed{0}$ ②② $\cos\left(\frac{\pi}{2}\right) = \cos 90^\circ = \boxed{0}$

②③ $\sin\left(\frac{3\pi}{2}\right) = \sin 270^\circ = \boxed{-1}$ ②④ $\cos \pi = \boxed{-1}$

Graphing Polar Equations Using Radian Units

Graph the following equations using radian units. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

• (25) $r = 1 + \cos \theta$



1st Point

$$\theta = 0$$

$$r = 1 + \cos(0)$$

$$r = 1 + 1$$

$$r = 2$$

$$(2, 0)$$

2nd Point

$$\theta = \frac{\pi}{2}$$

$$r = 1 + \cos\left(\frac{\pi}{2}\right)$$

$$r = 1 + 0$$

$$r = 1$$

$$\left(1, \frac{\pi}{2}\right)$$

3rd Point

$$\theta = \pi$$

$$r = 1 + \cos \pi$$

$$r = 1 - 1$$

$$r = 0$$

$$(0, \pi)$$

4th Point

$$\theta = -\frac{\pi}{3}$$

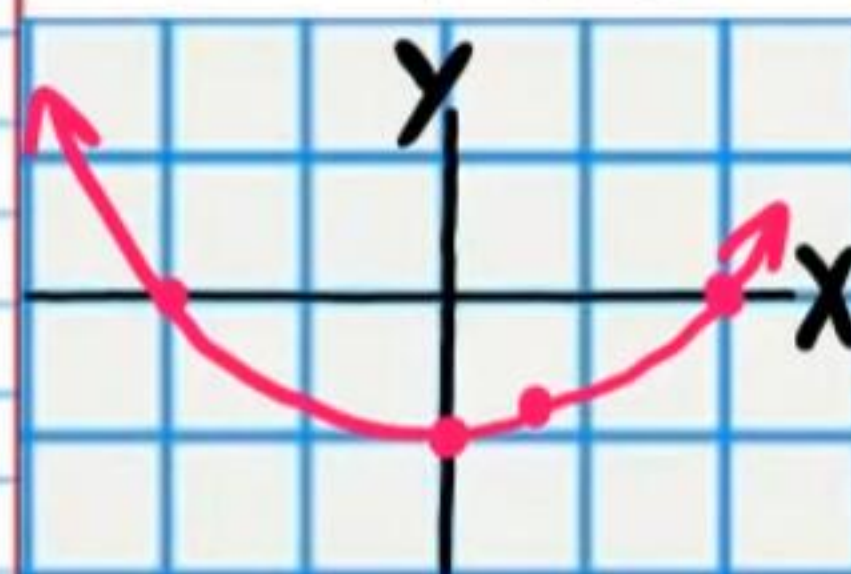
$$r = 1 + \cos\left(-\frac{\pi}{3}\right)$$

$$r = 1 + \frac{1}{2}$$

$$r = 1\frac{1}{2}$$

$$\left(1\frac{1}{2}, -\frac{\pi}{3}\right)$$

(26) $r = \frac{2}{1 - \sin \theta}$



1st Point

$$\theta = 0$$

$$r = \frac{2}{1 - \sin 0}$$

$$r = \frac{2}{1 - 0}$$

$$r = \frac{2}{1}$$

$$r = 2$$

$$(2, 0)$$

2nd Point

$$\theta = \pi$$

$$r = \frac{2}{1 - \sin \pi}$$

$$r = \frac{2}{1 - 0}$$

$$r = \frac{2}{1}$$

$$(2, \pi)$$

3rd Point

$$\theta = \frac{3\pi}{2}$$

$$r = \frac{2}{1 - \sin\left(\frac{3\pi}{2}\right)}$$

$$r = \frac{2}{1 - (-1)}$$

$$r = 1$$

$$\left(1, \frac{3\pi}{2}\right)$$

4th Point

$$\theta = -\frac{\pi}{3}$$

$$r = \frac{2}{1 - \sin\left(-\frac{\pi}{3}\right)}$$

$$r = \frac{2}{1 - \left(-\frac{\sqrt{3}}{2}\right)}$$

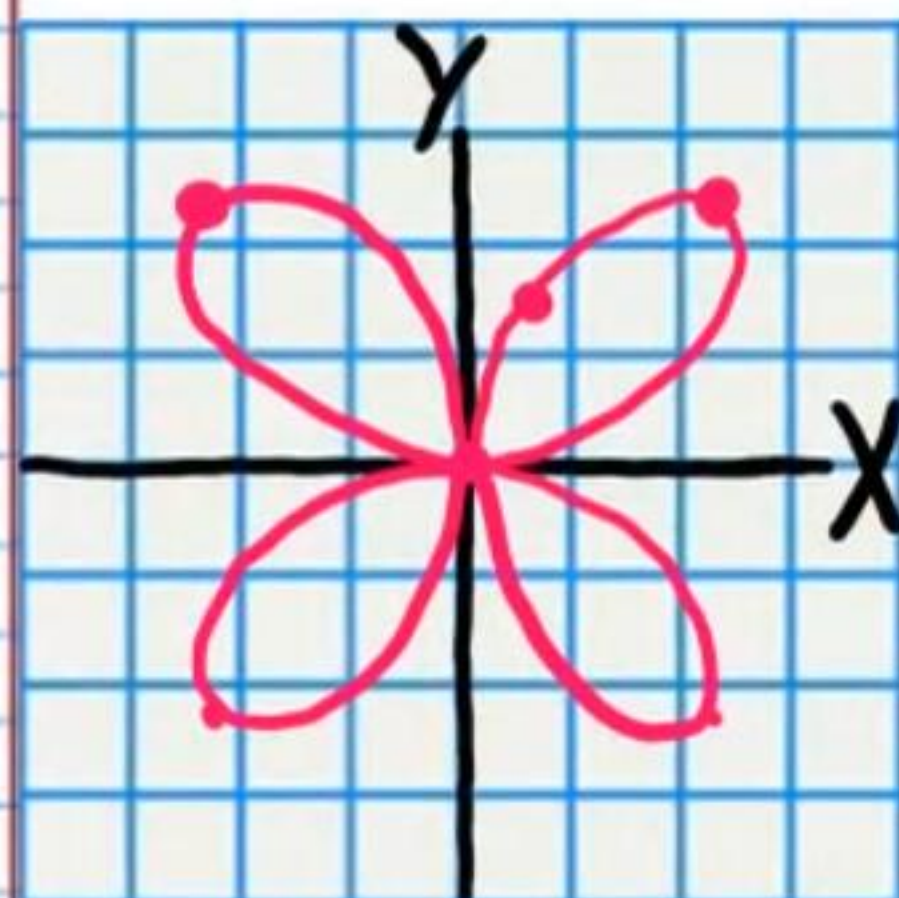
$$r = \frac{2}{1 + \frac{\sqrt{3}}{2}}$$

$$r = \frac{4}{2 + \sqrt{3}}$$

$$r \approx 1.071$$

$$\left(\frac{4}{2 + \sqrt{3}}, -\frac{\pi}{3}\right)$$

• (27) $r = 3\sin(2\theta)$



1st Point

$$\theta = 0$$

$$r = 3\sin(2(0))$$

$$r = 3\sin 0$$

$$r = 3(0)$$

$$r = 0$$

$$(0, 0)$$

2nd Point

$$\theta = \frac{\pi}{4}$$

$$r = 3\sin(2(\frac{\pi}{4}))$$

$$r = 3\sin(\frac{\pi}{2})$$

$$r = 3(1)$$

$$r = 3$$

$$(3, \frac{\pi}{4})$$

3rd Point

$$\theta = -\frac{\pi}{4}$$

$$r = 3\sin(2(-\frac{\pi}{4}))$$

$$r = 3\sin(-\frac{\pi}{2})$$

$$r = 3(-1)$$

$$r = -3$$

$$(-3, -\frac{\pi}{4})$$

4th Point

$$\theta = -\frac{\pi}{3}$$

$$r = 3\sin(2(-\frac{\pi}{3}))$$

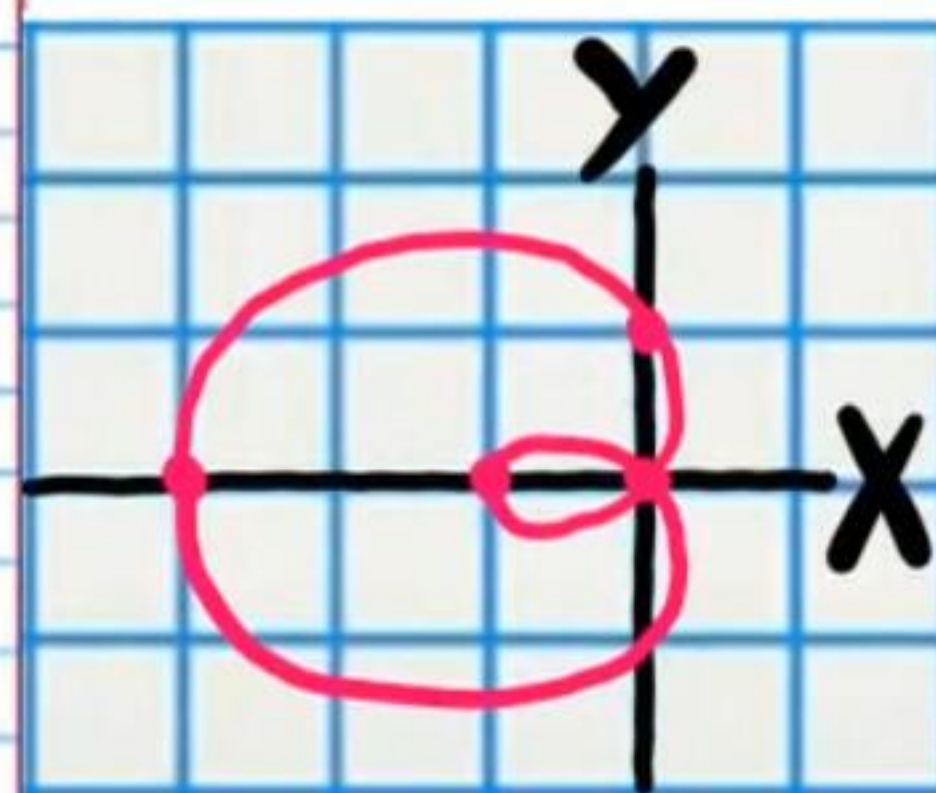
$$r = 3\sin(-\frac{2\pi}{3})$$

$$r = 3(-\frac{1}{2})$$

$$r = -1\frac{1}{2}$$

$$(-1\frac{1}{2}, -\frac{\pi}{3})$$

(28) $r = 1 - 2\cos\theta$



1st Point

$$\theta = 0$$

$$r = 1 - 2\cos 0$$

$$r = 1 - 2(1)$$

$$r = 1 - 2$$

$$r = -1$$

$$(-1, 0)$$

2nd Point

$$\theta = \frac{\pi}{2}$$

$$r = 1 - 2\cos(\frac{\pi}{2})$$

$$r = 1 - 2(0)$$

$$r = 1 - 2(0)$$

$$r = 1 - 0$$

$$r = 1$$

$$(1, \frac{\pi}{2})$$

3rd Point

$$\theta = \pi$$

$$r = 1 - 2\cos\pi$$

$$r = 1 - 2(-1)$$

$$r = 3$$

$$(3, \pi)$$

4th Point

$$\theta = \frac{\pi}{3}$$

$$r = 1 - 2\cos(\frac{\pi}{3})$$

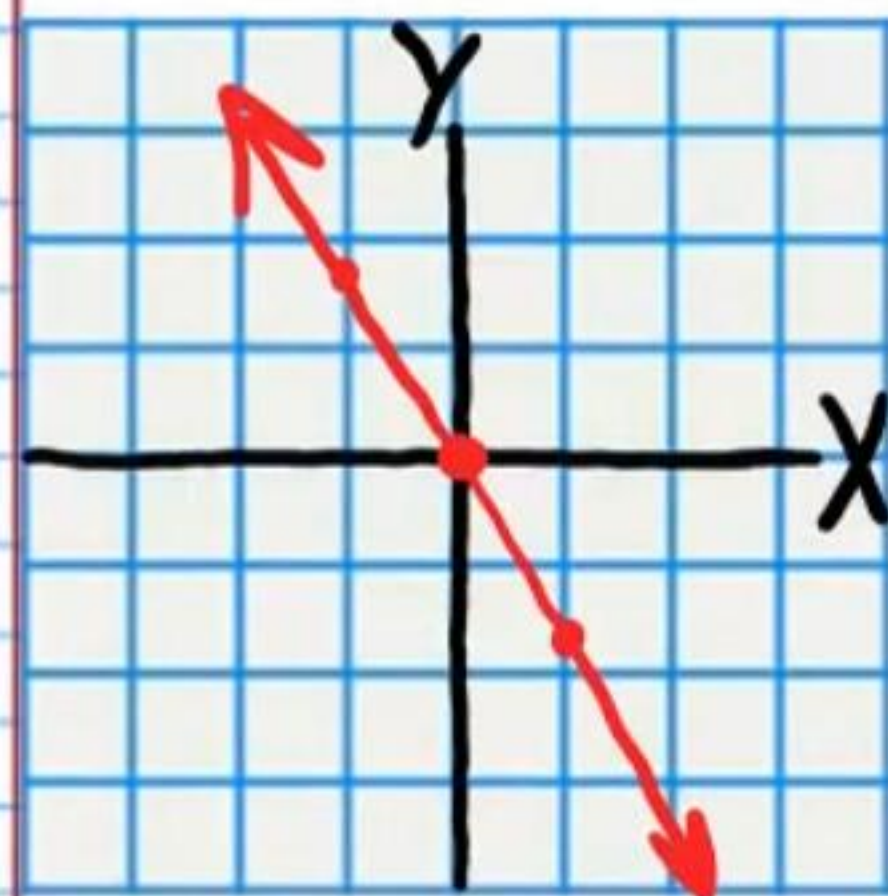
$$r = 1 - 2(\frac{1}{2})$$

$$r = 1 - 1$$

$$r = 0$$

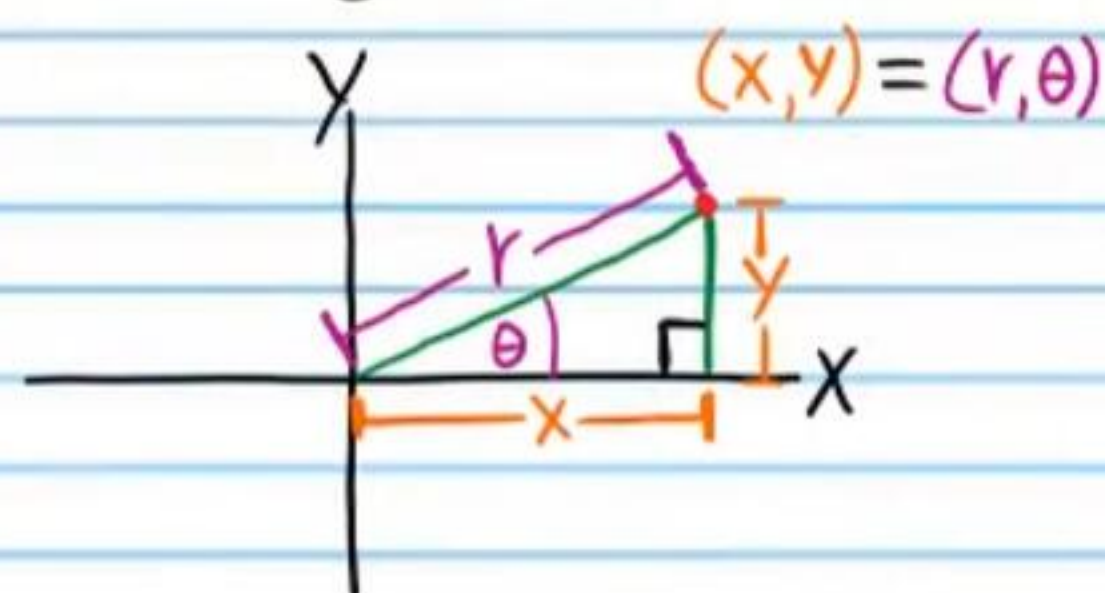
$$(0, \frac{\pi}{3})$$

• (29) $\theta = -\frac{4\pi}{3}$



Convert to rectangular form.

Use trigonometry



$$\tan \theta = \frac{y}{x}$$

$$\tan\left(-\frac{4\pi}{3}\right) = \frac{y}{x}$$

$$\tan(-240^\circ) = \frac{y}{x}$$

$$\frac{b\sqrt{3}}{-b} = \frac{y}{x}$$

$$-\sqrt{3} = \frac{y}{x}$$

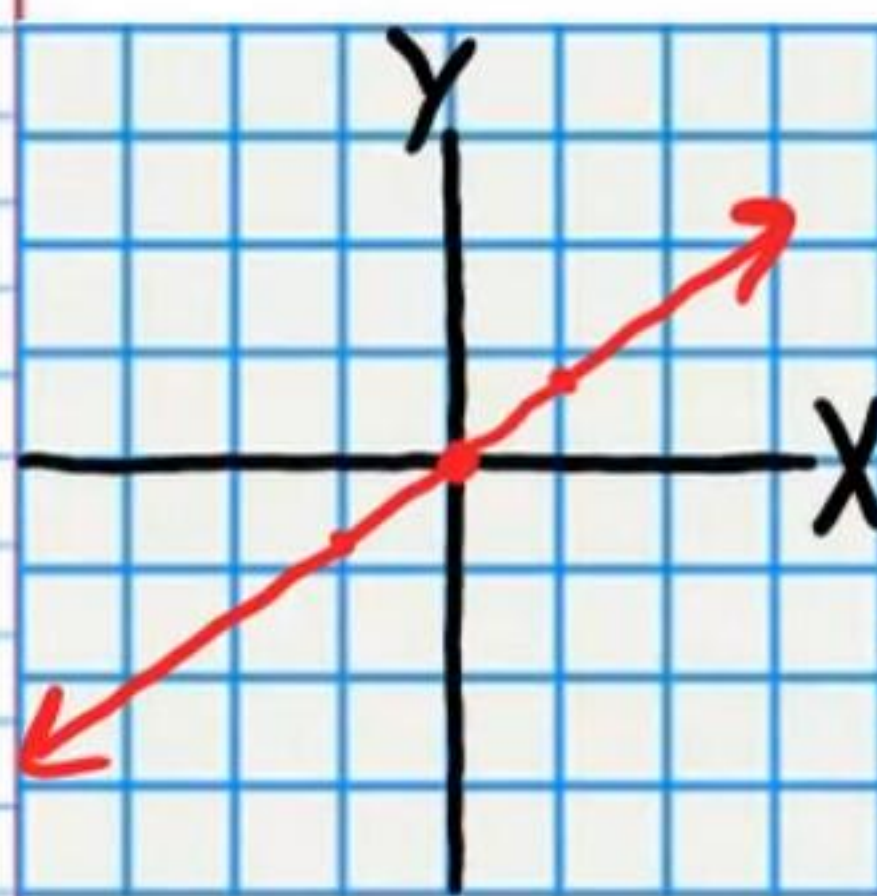
$$-\sqrt{3}x + 0 = y$$

$$mx + b = y$$

$$-1.732x + 0 = y$$

$$\frac{-1.732}{1}x + 0 = y$$

(30) $\theta = \frac{\pi}{5}$



$$\tan \theta = \frac{y}{x}$$

$$\tan \frac{\pi}{5} = \frac{y}{x}$$

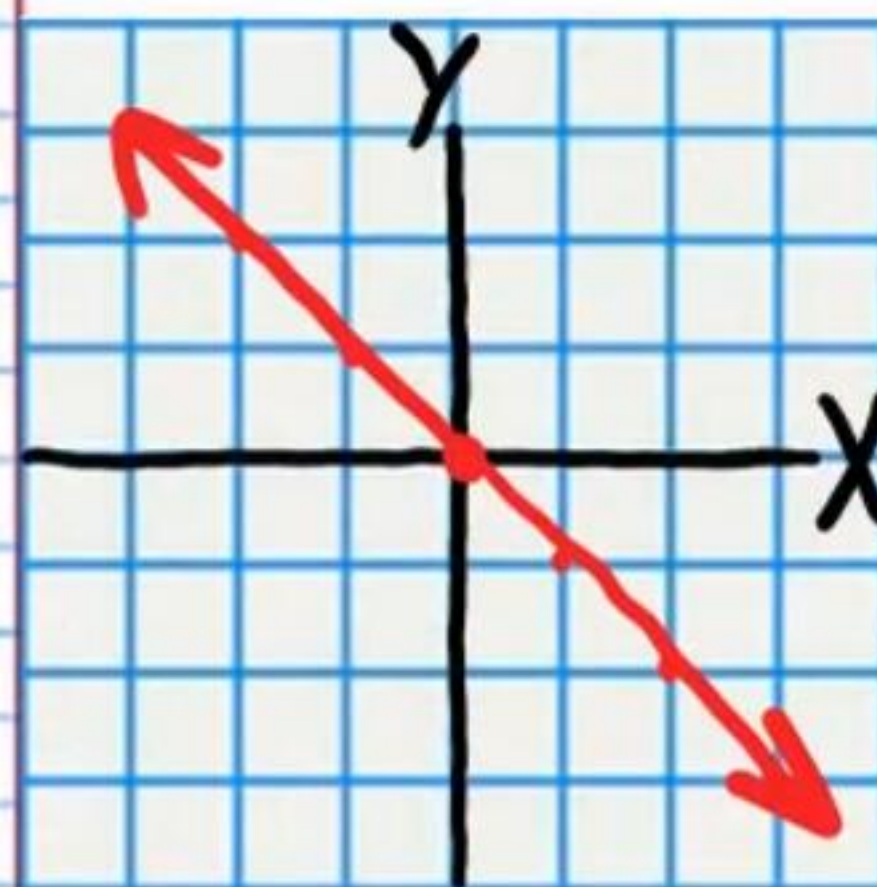
$$0.7265 = \frac{y}{x}$$

$$0.7265x = y$$

$$0.7265x + 0 = y$$

$$mx + b = y$$

• (31) $\theta = \frac{3\pi}{4}$



$$\tan \theta = \frac{y}{x}$$

$$\tan\left(\frac{3\pi}{4}\right) = \frac{y}{x}$$

$$\frac{b}{-b} = \frac{y}{x}$$

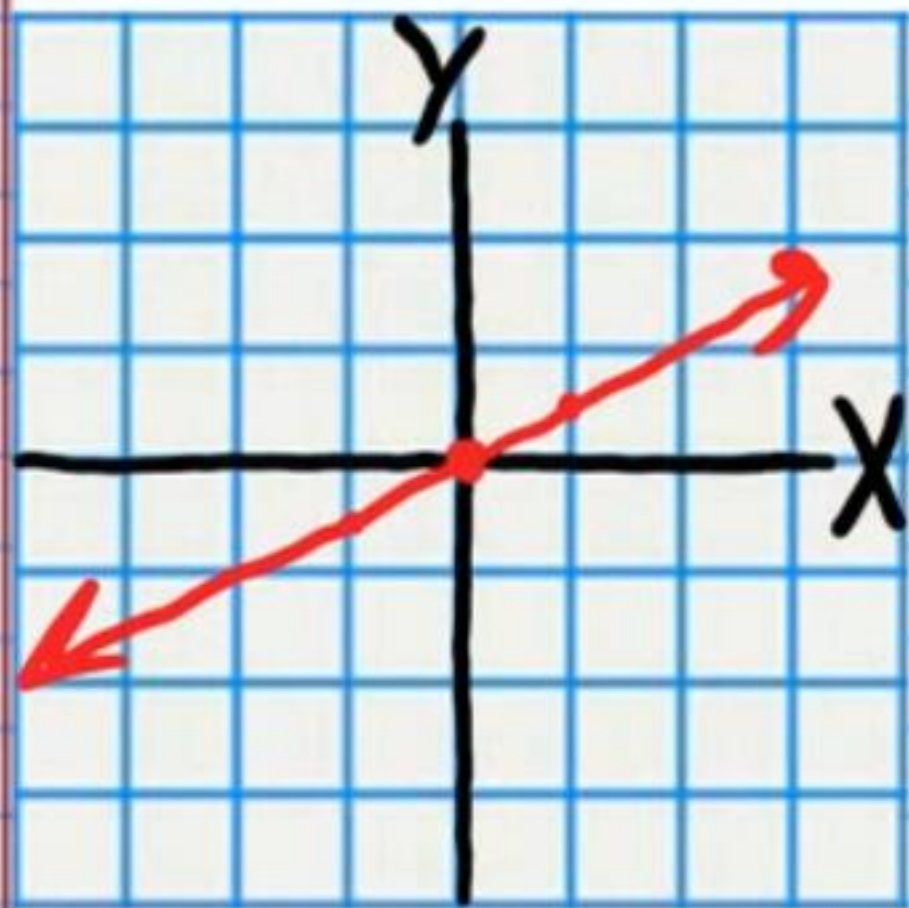
$$-1 = \frac{y}{x}$$

$$-x + 0 = y$$

$$\frac{-1}{1}x + 0 = y$$

$$mx + b = y$$

$$\textcircled{32} \theta = \frac{8\pi}{7}$$



$$\tan \theta = \frac{y}{x}$$

$$\tan\left(\frac{8\pi}{7}\right) = \frac{y}{x}$$

$$0.4815 = \frac{y}{x}$$

$$\frac{0.4815}{1}x + 0 = y$$

$$mx + b = y$$

Imaginary Numbers

$$i = \sqrt{-1}$$

$$i^2 = -1$$

Examples

$$i, -5i, 7i, i\sqrt{3}, \pi i, \frac{1}{2}i$$

Complex Numbers

Real Imaginary

$$a + bi$$

Examples

$$5 + 3i, -4 + i, \sqrt{7} - 2i, -9 - i\sqrt{8}, \frac{3}{4} + \frac{5}{7}i, \pi + 2i, 10, 6i, i$$

Simplify the expressions below.

$$\textcircled{33} (1 - 5i) + (-3 + 2i)$$

$$\boxed{-2 - 3i}$$

$$\textcircled{34} (6 + 3i) - (9 + i)$$

$$6 + 3i - 9 - i$$

$$\boxed{-3 + 2i}$$

$$\textcircled{35} -6i(-8 + 4i)$$

$$48i - 24i^2$$

$$48i - 24(-1)$$

$$48i + 24$$

$$\boxed{24 + 48i}$$

$$\textcircled{36} (2 + 5i)(4 - 7i)$$

$$8 - 14i + 20i - 35i^2$$

$$8 + 6i - 35(-1)$$

$$8 + 6i + 35$$

$$\boxed{43 + 6i}$$

$$\textcircled{37} (5 - 8i) + (7 - 4i)$$

$$\boxed{12 - 12i}$$

$$\textcircled{38} (-6 - 2i) - (-1 - 5i)$$

$$-6 - 2i + 1 + 5i$$

$$\boxed{-5 + 3i}$$

$$\textcircled{39} 10i(3 + i)$$

$$30i + 10i^2$$

$$30i + 10(-1)$$

$$\boxed{-10 + 30i}$$

$$\textcircled{40} (3 + i)(-8 + 2i)$$

$$-24 + 6i - 8i + 2i^2$$

$$-24 - 2i + 2(-1)$$

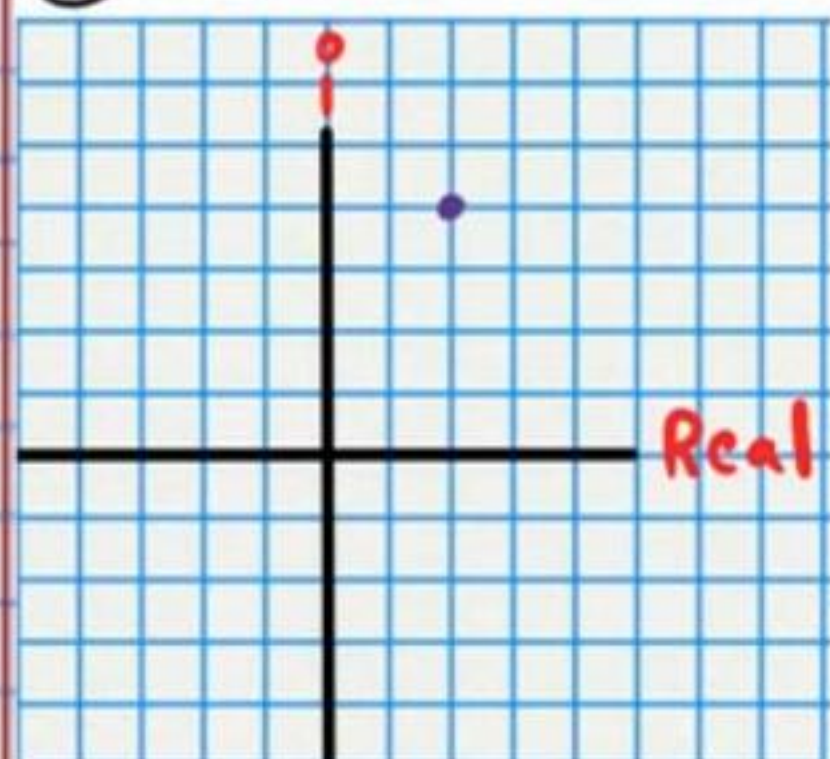
$$-24 - 2i - 2$$

$$\boxed{-26 - 2i}$$

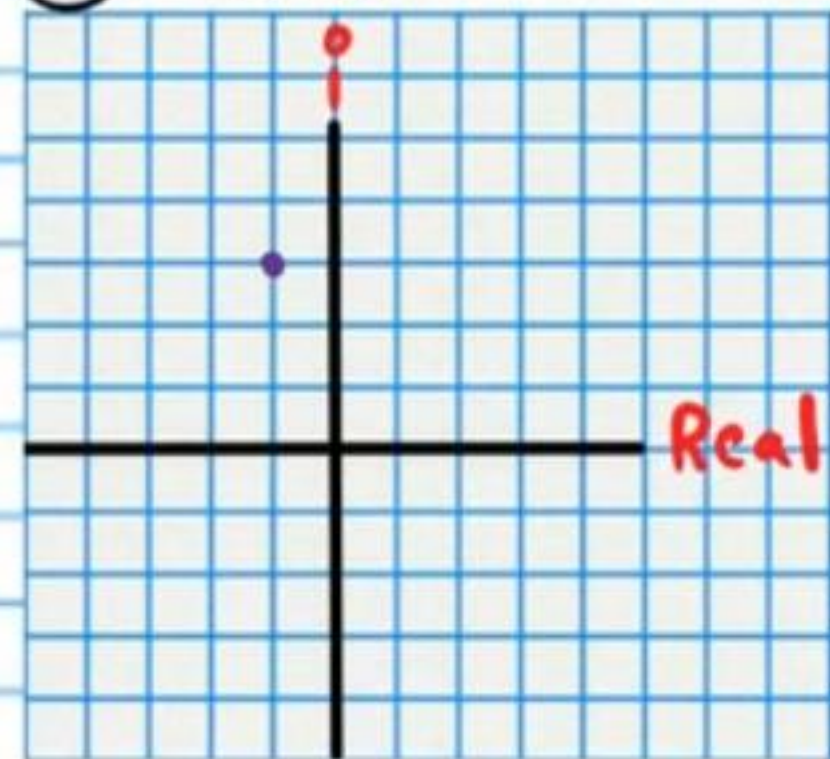
The Complex Plane

Graph the following complex numbers.

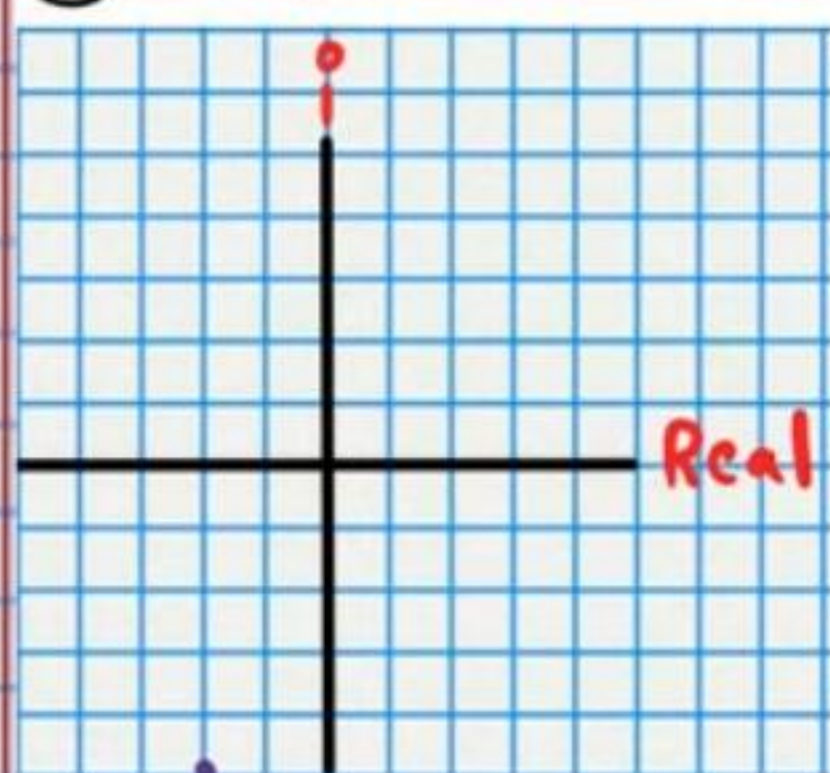
④① $2 + 4i$



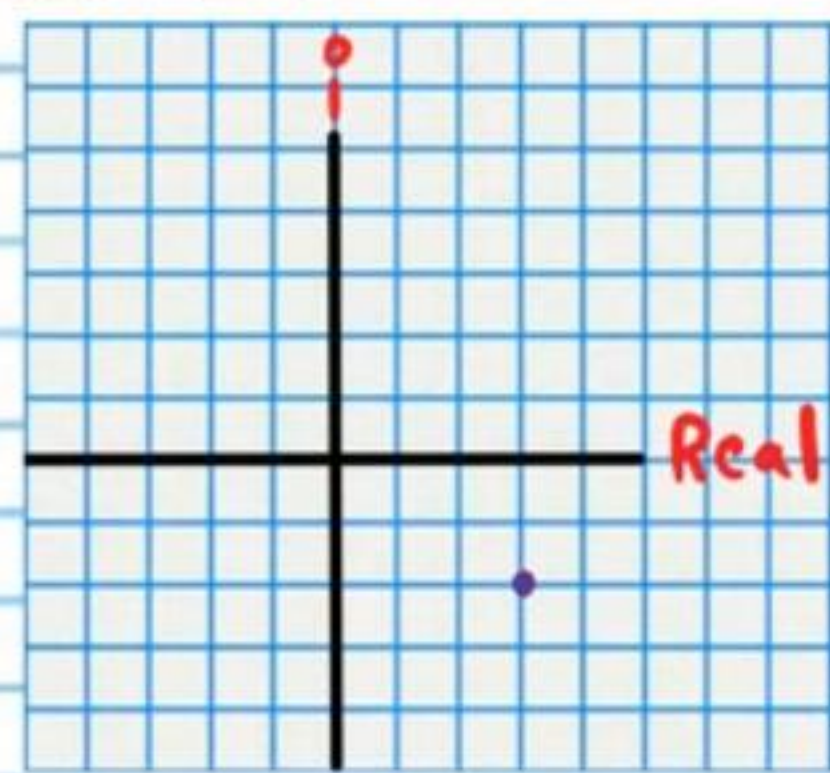
④② $-1 + 3i$



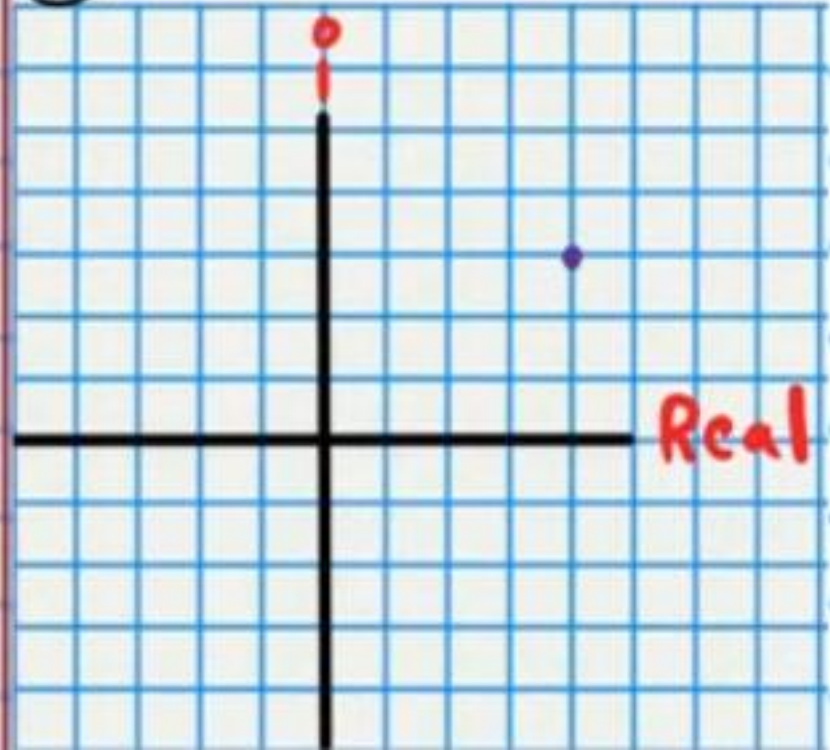
④③ $-2 - 5i$



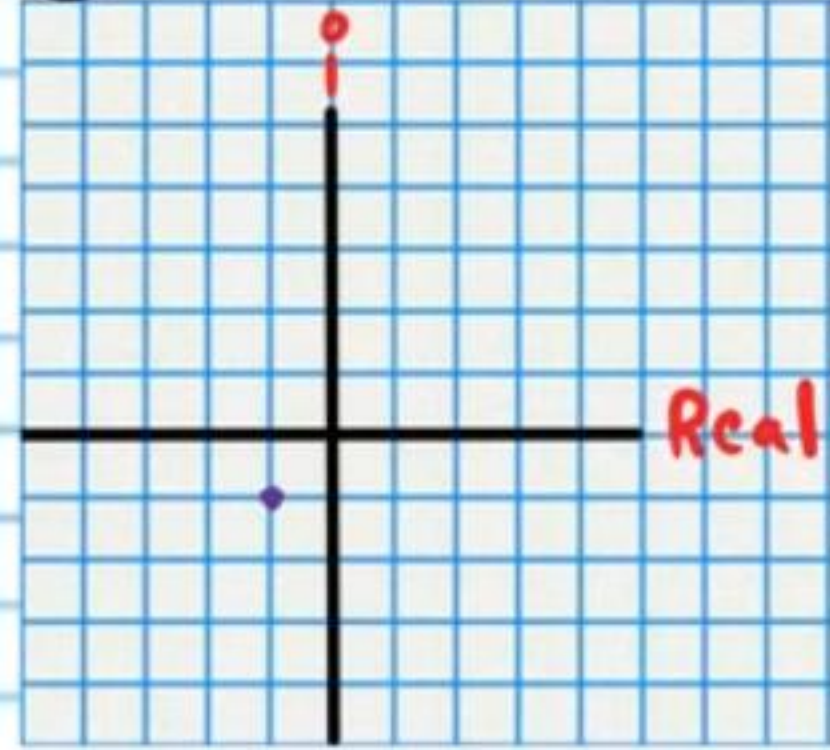
④④ $3 - 2i$



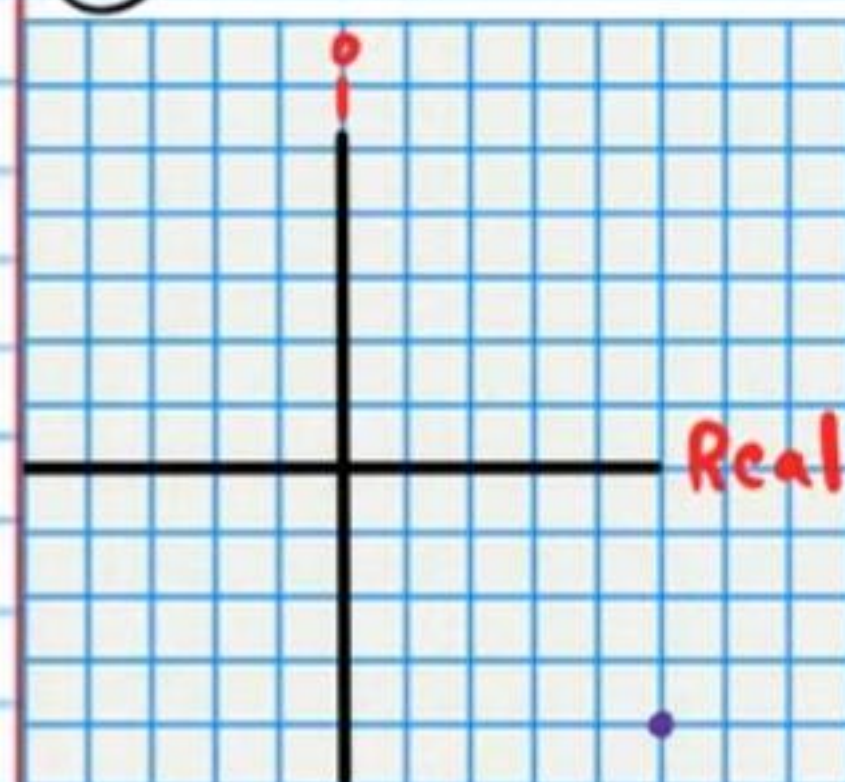
④⑤ $4 + 3i$



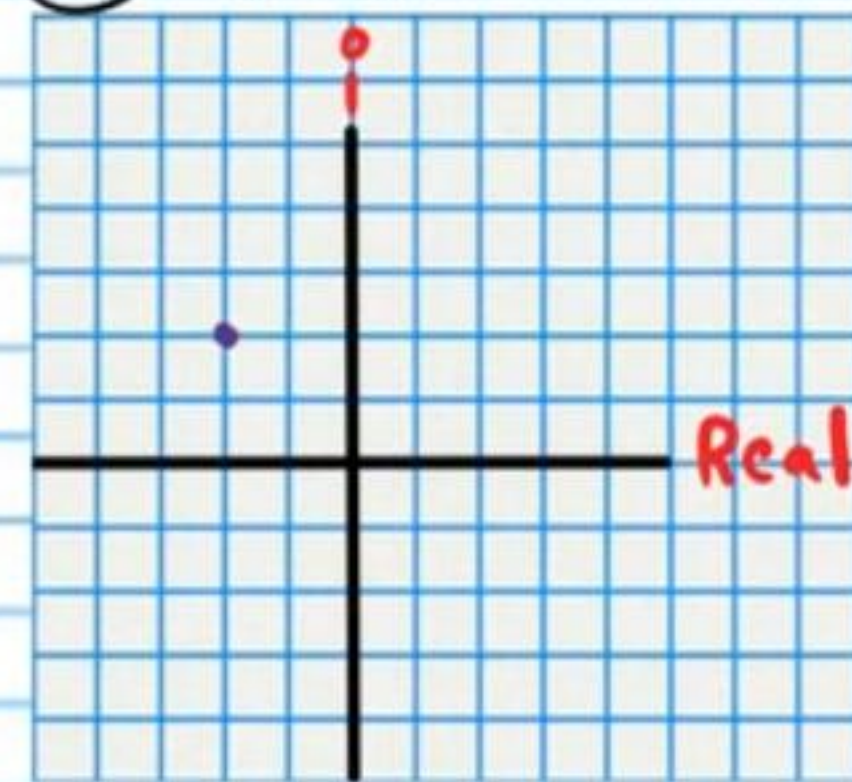
④⑥ $-1 - i$



④⑦ $5 - 4i$



④⑧ $-2 + 2i$



Jason Lisle
Mandelbrot set